

MINI PROJECT REPORT

LIBRARY MANAGEMENT SYSTEM

Based on Product Engineering & Design Thinking

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1. Introduction

The Library Management System is a software product designed to automate common academic library activities such as book cataloging, member management, book search, issue and return transactions, fine calculation, and administrative reporting. The project follows a product-engineering approach rather than treating coding as the complete development process.

2. Problem Statement

In a manual library, users may face delays while locating books, standing in queues for issue/return, and checking due dates. Librarians may also face difficulties maintaining accurate records, tracking overdue books, calculating fines, and preparing reports. The proposed system provides a centralized digital platform to reduce these operational difficulties.

3. Objectives

- Provide quick search and availability information for books.
- Maintain centralized records of books and library members.
- Automate book issue and return transactions.
- Calculate overdue fines automatically.
- Provide role-based access for students, faculty, librarians, and administrators.
- Generate basic reports for library administration.

4. Business Understanding

Core library operations considered in the project:

- Book Acquisition & Cataloging – maintain book details, authors, categories and shelf information.
- Membership Management – maintain student and faculty member records.
- Search & Circulation – search, issue, renew and return books.
- Financial Penalties – calculate fines for overdue returns.
- Administrative Analytics – provide basic usage and inventory reports.

5. System Development Life Cycle (SDLC)

The project follows these major lifecycle stages:

1. Requirement Gathering & Domain Analysis
2. Design Thinking & UX Wireframing
3. System Architecture & Database Schema Design
4. Implementation / Coding
5. Testing and CI/CD-oriented quality practices
6. Deployment, Monitoring and Continuous Improvement

6. Design Thinking

The project uses the five stages of human-centered Design Thinking.

6.1 Empathize

- Observe students, faculty members, librarians and administrators.

- Identify problems such as long queues, difficulty finding books, manual records and overdue tracking.

6.2 Define

Core problem statement: Library members experience delays in locating, reserving, borrowing and returning resources, while administrative staff lack centralized real-time visibility of library operations.

6.3 Ideate

- Online book search and availability.
- Automated due-date reminders.
- QR/barcode-based issue and return.
- Live inventory information.
- Automated fine calculation.

6.4 Prototype

- Login page
- Student/User search portal
- Librarian checkout dashboard
- Administrator reporting dashboard

6.5 Test

Test the prototype with representative users to evaluate navigation, clarity and task-completion flow before final implementation.

7. IBM Design Thinking

The project can be organized around the three IBM Design Thinking principles described in the reference material:

- Focus on User Outcomes – measure success by how easily users can find and reserve books.
- Restless Reinvention – continuously improve the platform using usage data and feedback.
- Diverse Empowered Teams – involve product, UX, development, QA, database and operations roles.

8. IBM Loop: Observe → Reflect → Make

Observe: study user interactions, system logs and operational friction.

Reflect: analyze metrics, feedback and priorities.

Make: improve the UI, code, automation and deployment process.

9. User Personas

Persona	Objective	Pain Point	System Feature
Student	Find academic books quickly	Queues and unexpected fines	Search portal and due-date alerts
Librarian	Process issue/return quickly	Manual records and stock discrep	nlscsieuse/return interface and fine tracker
Administrator	Monitor library usage	Manual report preparation	Analytics and reporting dashboard

10. Stakeholder Matrix

Stakeholder Group	Role / Responsibility
Business Stakeholders	College management, library administration, students and faculty; define business needs and validate
Engineering Stakeholders	Developers, UI/UX designers and QA; build the application and ensure quality.

Stakeholder Group	Role / Responsibility
Operations Stakeholders	DevOps, database and security roles; maintain infrastructure, database integrity and deployment automation

11. Functional Requirements

FR-01 – Authentication: Role-based login for Admin, Librarian, Student and Faculty.

FR-02 – Book Catalog: Add, view, update and delete book, author, category and shelf details.

FR-03 – Circulation: Issue and return books while updating availability.

FR-04 – Fine Engine: Calculate overdue fines automatically.

FR-05 – Notifications: Provide due-date and overdue reminders.

12. Non-Functional Requirements

- Security – protect accounts and user data using secure authentication and encrypted communication.
- Performance – book searches should return results quickly under normal concurrent usage.
- Availability & Scalability – the system should support growth in users and records.
- Maintainability – use modular code and clear separation of application components.

13. Product Vision

To deliver a modern and automated Library Management System that reduces manual administrative work, streamlines book circulation, and provides an easy digital catalog experience for an academic community.

14. MVP (Minimum Viable Product) Scope

- User Login and Profile with role-based access.
- Book Search and Availability Status.
- Basic Issue and Return Transaction Management.
- Automated Fine Calculation.
- Basic Administrative Reporting.

15. System Modules

Login & Authentication: Validates users and provides role-based access.

Book Management: Maintains book title, author, category, ISBN and shelf details.

Member Management: Maintains student and faculty profiles.

Search Module: Allows users to search books and check availability.

Issue Book: Records book issue and due date.

Return Book: Records return and updates inventory.

Fine Management: Calculates overdue penalties.

Reports: Provides inventory, circulation and fine reports.

16. Proposed System Architecture

User Interface → Application / Business Logic → Database

Layer	Components
Presentation Layer	Login, Search, Issue/Return, Dashboard, Reports
Application Layer	Authentication, Catalog, Circulation, Fine and Notification logic
Data Layer	Users, Books, Transactions, Fines, Categories and Authors tables

17. Database Design

Suggested main entities:

Table	Important Fields
User	User_ID, Name, Email, Role, Password
Book	Book_ID, Title, ISBN, Author_ID, Category_ID, Shelf, Quantity, Available
Author	Author_ID, Author_Name
Category	Category_ID, Category_Name
Transaction	Transaction_ID, User_ID, Book_ID, Issue_Date, Due_Date, Return_Date, Status
Fine	Fine_ID, Transaction_ID, Amount, Payment_Status

18. Use Case Description

- Student/Faculty: Login → Search Book → Check Availability → Issue/Reserve → Return Book → View Fine.
- Librarian: Login → Manage Books → Issue/Return → Manage Members → Check Fines → Generate Reports.
- Administrator: Login → Manage Users/Roles → View Analytics → Generate Reports.

19. Agile Sprint Plan

Sprint	Main Activities
Sprint 1	Requirements, personas, problem definition and UI wireframes
Sprint 2	Database design, login and book catalog
Sprint 3	Search, issue/return and fine calculation
Sprint 4	Reports, testing, feedback and MVP refinement

20. Testing

- Unit Testing – test individual functions such as login validation and fine calculation.
- Integration Testing – verify communication between UI, application logic and database.
- System Testing – test complete issue, return, search and reporting workflows.
- User Acceptance Testing – validate usability with representative library users.

21. Future Scope

- Mobile application support.
- QR/barcode-based self-checkout.
- Automated email/SMS notifications.
- Advanced analytics dashboard.
- AI-based book recommendation as a future enhancement.

22. Conclusion

The Library Management System applies product engineering and Design Thinking to a practical academic library problem. By combining centralized data, automated circulation, fine calculation, user-focused design and iterative testing, the system can reduce manual effort and improve the overall library workflow.

23. Reference Basis

This mini-project structure is based on the uploaded training document on Library Management Platform, especially its sections on software product development, library business understanding, Design Thinking, IBM Design Thinking, personas, stakeholder matrix, requirements, MVP scope, roadmap and architecture. ■filecite■turn0file0■L97-L109■